Evaluating Locally Discriminative Learning for Hallucination Suppression in Remote-Sensing Super-Resolution

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***Abstract*—While GAN-based super-resolution methods can synthesize visually sharp textures, they often introduce hallucinated high-frequency artifacts that lack structural reliability. This problem is particularly critical in remote-sensing imagery, where geometric fidelity is essential for accurate downstream analysis. This work evaluates the use of locally discriminative learning as an artifact-aware supervision strategy for remote-sensing super-resolution using the UC Merced Land Use dataset. When locally discriminative supervision is applied within an ESRGAN-based framework under identical training conditions, reconstruction quality improves substantially compared with a standard ESRGAN baseline [3]. Specifically, Peak Signal-to-Noise Ratio increases from 21.95 dB to 26.18 dB, Structural Similarity Index improves from 0.565 to 0.741, and Learned Perceptual Image Patch Similarity decreases from 0.400 to 0.241. Comparisons with non-adversarial and transformer-based methods further show that architectural changes alone are insufficient to suppress hallucinated artifacts. Qualitative results confirm reduced grid-like distortions in urban regions and more stable texture patterns in agricultural areas, indicating that locally discriminative, variance-aware supervision is effective for artifact-sensitive remote-sensing super-resolution.**

***Keywords*—Deep learning, generative adversarial networks, image super-resolution, locally discriminative learning**

I. Introduction

Single-image super-resolution aims to reconstruct a higher-resolution image from a single degraded observation and is a long-standing problem in image analysis and computer vision research [6], [8]. Recent progress in generative adversarial network–based approaches has led to substantial improvements in perceptual quality, with methods such as ESRGAN producing visually sharp and detailed reconstructions [10], [13]. However, adversarial optimization is known to introduce hallucinated high-frequency details that are not present in the original scene, reflecting the perception–distortion trade-off inherent in super-resolution [1]. While such hallucinations may be visually acceptable for photographic enhancement, they are problematic in scientific imaging domains where accuracy and reliability are critical. Remote-sensing imagery requires high geometric fidelity and structural consistency due to its strong spatial regularity. Even small distortions can alter man-made structures, disrupt repetitive agricultural patterns, and degrade downstream tasks such as land-use classification, change detection, and scene interpretation [11], [15]. Super-resolution methods for Earth observation must therefore balance perceptual sharpness with structural reliability. Prior efforts to mitigate adversarial artifacts show trade-offs: removing adversarial loss improves numerical fidelity but often produces oversmoothed results with limited realism [6], [7], while transformer-based models such as SwinIR [2] perform well on natural images, recent diffusion-based approaches have also demonstrated strong reconstruction capability for image super-resolution [4]. However, diffusion-based methods often involve higher computational cost and more complex inference pipelines, which may limit their practicality in certain remote-sensing applications. These findings indicate that architectural changes or loss design alone are insufficient to prevent hallucinated details in remote-sensing super-resolution.

Locally discriminative learning was introduced as an artifact-aware supervision strategy designed to suppress GAN-induced hallucinations while preserving legitimate high-frequency details [1]. Rather than applying uniform reconstruction penalties, this approach identifies unstable regions through residual variance analysis and selectively constrains them using exponential moving average–guided refinement. Although locally discriminative learning (LDL) has demonstrated effectiveness on natural-image datasets, its applicability to remote-sensing imagery, where spatial regularity, repetitive structures, and geometric accuracy are essential, has not been thoroughly evaluated. This gap motivates the present study.

This work evaluates the application of locally discriminative learning to remote-sensing super-resolution using the UC Merced Land Use dataset [12]. By comparing performance against an ESRGAN baseline, a non-adversarial variant, and a transformer-based model under identical evaluation settings, this study examines whether locally discriminative, variance-aware supervision can effectively reduce hallucinated artifacts while preserving structural fidelity. The findings provide evidence for the effectiveness of LDL-style mechanisms in artifact-sensitive super-resolution and support further evaluation across additional imaging domains.

II. Method Background and Adaptation

This study builds on the ESRGAN super-resolution framework together with the locally discriminative learning (LDL)

strategy. The objective is not to redesign the model architecture, but to examine how artifact-aware supervision affects reconstruction behavior in structured remote-sensing imagery.

Real-ESRGAN builds upon the ESRGAN framework by refining the Residual-in-Residual Dense Block RRDB architecture and improving training stability under real-world degradations [10]. By combining adversarial training with perceptual loss, the model produces visually sharp reconstructions. However, the adversarial objective can also encourage the synthesis of high-frequency content that is not present in the original scene, particularly in regions with strong geometric regularity.

LDL was introduced to mitigate this issue by guiding the network to focus correction on regions exhibiting unstable reconstruction behavior rather than applying uniform penalties across the entire image [1]. The method begins by examining the residual between the reconstructed image and its ground truth. Let I_{HR} denote the reference high-resolution image and I_{SR} the generated output. The residual is defined as

$$R = I_{HR} - I_{SR}. \tag{1}$$

Local variance is then computed over sliding windows of R to identify areas where reconstruction errors fluctuate significantly. Because naturally textured regions can also exhibit high variance, LDL introduces a global scaling factor

$$\sigma = (Var(R))^{1/a}, \quad a = 5, \tag{2}$$

which moderates the strength of penalization according to the overall difficulty of the image.

To further distinguish true artifacts from normal detail, LDL maintains an exponential moving average (EMA) version of the generator alongside the actively updated model. Let R_1 and R_2 represent residuals from the current generator Ψ and the EMA generator Ψ_{EMA}. A pixel at location (i, j) is treated as unstable when

$$|R_1(i, j)| \geq |R_2(i, j)|. \tag{3}$$

This comparison allows the EMA model to act as a stable reference so that only pixels showing degraded behavior are selected for additional supervision.

The final objective augments the standard adversarial loss with a targeted artifact term,

$$L_{LDL} = L_{GAN} + \beta L_{artifact}, \tag{4}$$

where β controls the contribution of the artifact-aware constraint. By directing optimization toward problematic regions instead of uniformly smoothing the image, LDL reduces hallucinated structures while maintaining meaningful high-frequency information.

III. Application to Remote-Sensing Super-Resolution

Remote-sensing imagery imposes strict requirements on geometric fidelity because of its strong spatial regularity and sensitivity to distortion. Hallucinated high-frequency details produced by adversarial super-resolution can modify man-made structures, disrupt repetitive layouts, and reduce the reliability of downstream geospatial analysis. Such scenes can be broadly categorized into three region types: smooth or large-scale areas (Type A), naturally irregular textures (Type B), and fine-scale structured regions (Type C), including dense building grids, roof patterns, and thin linear features. Type C regions are particularly vulnerable to GAN-induced artifacts, where adversarial optimization may exaggerate repetitive structures or introduce aliasing effects. LDL is well suited to this setting because it applies supervision selectively to unstable regions identified through residual variance and EMA-based refinement. This targeted correction suppresses artifact-prone responses while preserving legitimate detail in stable or naturally textured areas.

In this study, LDL is integrated into the ESRGAN framework for remote-sensing super-resolution using the UC Merced Land Use dataset. No domain-specific architectural changes are introduced. Performance is compared with a standard ESRGAN baseline, a non-adversarial variant, and a transformer-based model under identical conditions to evaluate whether variance-aware supervision improves structural fidelity and reduces hallucinated artifacts in aerial imagery.

IV. Experimental Design and Implementation

This section describes the experimental setup for evaluating locally discriminative learning (LDL) in remote-sensing super-resolution. The pipeline follows prior ESRGAN and LDL implementations, with additional reference models evaluated under identical conditions to ensure a fair and reproducible comparison. *The full implementation is publicly available at https://github.com/Sarah-Wai/LDL_UC-Merced.*

A. Dataset Preparation and Training Configuration

Experiments use the UC Merced Land Use dataset [12], containing 2,100 RGB images (256×256) across 21 categories. Low-resolution inputs are generated via $\times 4$ bicubic downsampling to 64×64, with a 70%/15%/15% split for training, validation, and testing.

Low-resolution images are upsampled to 256×256 to match ESRGAN input. During training, paired patches (128×128) are randomly cropped, with data augmentation including flips, rotations, and mild HSV jitter.

All ESRGAN-based models (baseline, NoGAN, and LDL-SR) use identical settings: Adam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$), batch size 16, and learning rate 1×10^{-4} with cosine decay over $\sim$50k iterations. The NoGAN variant removes adversarial loss, while LDL-SR incorporates residual analysis and EMA refinement ($\beta = 1$, $\alpha = 0.999$). Experiments are conducted on a single NVIDIA T4 GPU (16 GB), and SwinIR is evaluated using pretrained weights as a reference model.

B. Training Stability

Training stability is illustrated in Fig. 1. The generator loss curves show smooth convergence over 50,000 iterations, with pixel-level and artifact-related terms remaining within

a narrow range. Perceptual and adversarial losses stabilize without large oscillations, indicating controlled adversarial optimization. This behavior is consistent with stabilization effects reported in previous LDL studies [1].

Discriminator losses exhibit the expected fluctuations characteristic of adversarial optimization [10], but remain bounded throughout training and show no evidence of divergence or mode collapse. These observations indicate stable optimization behavior under the adopted training settings.

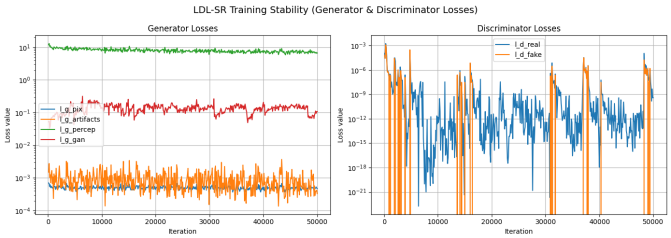

Fig. 1. Training behavior over 50,000 iterations. Generator losses converge smoothly, while discriminator losses remain bounded despite expected adversarial fluctuations.

C. Evaluation Metrics and Implementation Fidelity

Model performance was assessed using a combination of fidelity-based and perceptual metrics. The peak Signal-to-Noise Ratio was used to measure the pixel-level reconstruction accuracy and was calculated on the luminance channel of the YCbCr color space [8]. The structural Similarity Index was used to evaluate structural consistency and luminance preservation, also on the luminance channel [8]. Learned Perceptual Image Patch Similarity was employed as a perceptual metric using pretrained deep feature extractors. All reported values represent averages over the held-out test set and follow evaluation protocols used in prior ESRGAN and LDL studies.

The experimental pipeline was implemented to closely follow the specifications described in prior LDL work. The ESRGAN backbone, RRDB architecture, perceptual loss formulation, and LDL-related procedures, including artifact map computation, variance normalization, and EMA-guided refinement, were reproduced according to published descriptions.

Implementation correctness was verified through inspection of intermediate outputs, including residual maps and EMA-based predictions, as well as through repeated training runs with different random seeds. Training logs were monitored to confirm stable optimization behavior. These steps ensured that the experimental comparison was conducted in a consistent and reliable manner.

V. Results, Analysis, and Insights

The proposed LDL-SR framework is evaluated on the UC Merced land-use dataset. Qualitative examples are first presented to illustrate LDL-SR behavior across representative scene types, followed by an integrated interpretation of these observations. Quantitative ablation studies and comparisons with reference models are then provided to support the visual and analytical findings.

A. Qualitative Observations Across Land-Use Categories

This subsection presents representative qualitative results corresponding to the three canonical region types defined in the LDL framework: smooth regions (Type A), natural random textures (Type B), and fine-scale aliasing structures (Type C) [1]. These examples illustrate how LDL-SR behaves across diverse land-use categories commonly found in remote-sensing imagery.

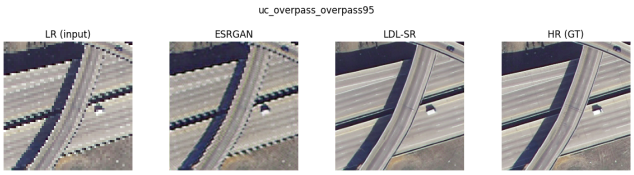

Fig. 2. Urban overpass comparison.

Urban overpass scenes (Fig. 2): Urban overpasses are dominated by thin structural edges, junctions, and repetitive linear patterns, which closely correspond to Type C regions [1]. ESRGAN introduces grid-like noise and jagged edges along bridge rails and concrete boundaries due to adversarial overshoot [10]. In contrast, LDL-SR suppresses these artifacts, restoring smoother contours and more coherent linear geometry.

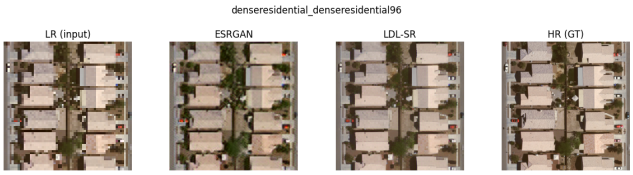

Fig. 3. Residential scene comparison.

Residential scenes (Fig. 3): Residential areas typically contain a mixture of structured rooftops and surrounding vegetation, mapping primarily to Type B regions [1]. ESRGAN tends to exaggerate roof textures and vegetation, often producing repeated or unstable patterns not present in the ground truth [10]. LDL-SR generates more spatially consistent roof surfaces while reducing spurious high-frequency detail in vegetation.

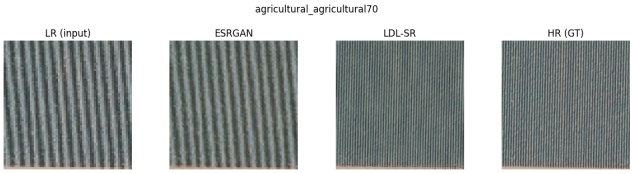

Fig. 4. Agricultural field comparison.

Agricultural field scenes (Fig. 4): Agricultural fields exhibit repetitive geometric layouts and regular spacing, making them particularly sensitive to aliasing artifacts characteristic of Type C regions [1]. ESRGAN frequently introduces ringing artifacts or misaligned crop-row spacing [10]. LDL-SR preserves consistent linear patterns, suppresses aliasing, and reconstructs more faithful agricultural structures.

Runway scenes (Fig. 5): Runway imagery is dominated

by engineered linear infrastructure with sharp boundaries, strongly aligned with Type C regions [1]. ESRGAN often produces fragmented lane markings, irregular edge transitions, and unstable asphalt textures, reflecting the perception–distortion trade-off inherent in GAN-based SR [5], [9]. LDL-SR significantly reduces these distortions by enforcing cleaner edge continuity, smoother runway boundaries, and more uniform surface appearance.

Fig. 5. Runway scene comparison.

Across all examples, LDL-SR consistently reduces hallucinated high-frequency artifacts while preserving meaningful structural detail. Improvements are most pronounced in scenes dominated by Type C regions, where aliasing and adversarial instability are prevalent [1], [6]. Scenes containing natural stochastic textures (Type B) primarily benefit from reduced artificial graininess, while smooth regions (Type A) remain largely unchanged, indicating that LDL avoids unnecessary suppression in stable areas. This selective behavior aligns with the design objective of LDL and explains its robustness across diverse land-use categories.

B. Ablation Study and Comparison With Reference Models

Table I presents an ablation study evaluating the contribution of individual components within the LDL framework on the UC Merced test set. Performance is evaluated using three standard super-resolution metrics: Peak Signal-to-Noise Ratio (PSNR), which measures pixel-level reconstruction fidelity in decibels (dB); Structural Similarity Index Measure (SSIM), which assesses structural consistency between reconstructed and reference images [8]; and Learned Perceptual Image Patch Similarity (LPIPS), which quantifies perceptual distance using deep feature representations [14].

The ESRGAN baseline achieves 21.9455 dB PSNR, 0.5650 SSIM, and 0.3607 LPIPS. Removing the LDL mechanism and introducing only the variance-based artifact map (M) leads to a clear degradation in performance, with PSNR decreasing by 4.1%, SSIM by 16.6%, and LPIPS increasing by 23.0%. This behavior indicates that naïve penalization of high-variance regions suppresses both artifacts and genuine high-frequency details, rather than discriminating between them [1]. Applying global variance normalization (σM) partially recovers PSNR, with only a 0.06% decrease relative to ESRGAN, but still degrades SSIM by 4.1% and increases LPIPS by 24.8%, demonstrating that global scaling alone is insufficient for effective artifact–detail discrimination [1]. Incorporating EMA-based refinement (M_{refine}) improves training stability but remains inferior to the baseline, with PSNR and SSIM still reduced by 3.4% and 3.5%, respectively. These results show that removing either stable residual-variance estimation or refinement leads to consistent performance degradation, confirming that individual LDL components are insufficient when applied in isolation [1].

In contrast, the full LDL-SR model achieves 26.1798 dB PSNR, 0.7407 SSIM, and 0.2127 LPIPS. Compared to ESRGAN, this corresponds to a 19.3% improvement in PSNR, a 31.1% increase in SSIM, and a 41.0% reduction in LPIPS. These gains clearly demonstrate that LDL is essential for simultaneously improving reconstruction fidelity and perceptual quality. Removing any component of LDL results in a measurable and consistent decline across all evaluation metrics, confirming that stable residual-variance estimation, EMA-guided refinement, and selective loss weighting must operate jointly to obtain the full benefit of LDL [1].

TABLE I
ABLATION RESULTS ON THE UC MERCED TEST SET. BEST RESULTS ARE SHOWN IN **BOLD**.

Model	PSNR↑	SSIM↑	LPIPS↓
ESRGAN	21.9455	0.5650	0.3607
ESRGAN + M	21.0528	0.4715	0.4435
ESRGAN + σM	21.9319	0.5420	0.4502
ESRGAN + M_{refine}	21.2039	0.5452	0.4442
LDL-SR (full)	**26.1798**	**0.7407**	**0.2127**

Although formal statistical tests such as paired t-tests could not be conducted due to missing per-image evaluation logs, the observed improvements remain consistent and substantial across all metrics. LDL-SR achieves the highest PSNR and SSIM and the lowest LPIPS among all ablation variants. Each partial configuration shows a clear performance decline relative to the full model, providing strong empirical evidence for the effectiveness of all LDL components.

Table II compares LDL-SR with additional reference models. Removing adversarial loss (NoGAN) improves PSNR and SSIM over ESRGAN by 6.6% and 10.2%, respectively, confirming that adversarial optimization can introduce instability in structured aerial imagery [6]. However, NoGAN remains substantially inferior to LDL-SR, with PSNR and SSIM still 10.8% and 15.8% lower, indicating that artifact suppression alone is insufficient for high-quality reconstruction [1]. The transformer-based SwinIR model [2], evaluated in inference-only mode, underperforms ESRGAN with a 3.6% decrease in PSNR, suggesting limited generalization of natural-image super-resolution architectures to remote-sensing imagery without domain-specific adaptation. LDL-SR consistently achieves the best performance across all evaluation metrics.

TABLE II
COMPARISON OF LDL-SR WITH REFERENCE MODELS ON UC MERCED TEST SET. BEST RESULTS IN **BOLD**.

Model	PSNR (dB)↑	SSIM↑	LPIPS↓
ESRGAN (baseline)	21.9455	0.5650	0.3607
NoGAN (no adversarial loss)	23.3947	0.6228	0.3716
SwinIR (inference-only)	21.1540	0.5910	0.3685
LDL-SR (Ours)	**26.1798**	**0.7407**	**0.2127**

C. Artifact–Detail Discrimination Analysis

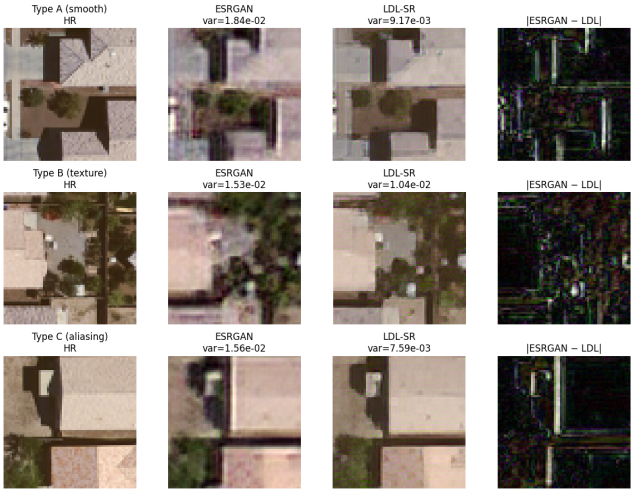

Fig. 6. Artifact–detail discrimination across three region types (Type A, B, and C). For each row: HR reference, ESRGAN output with residual variance, LDL-SR output with residual variance, and absolute difference map $|ESRGAN - LDL|$.

Fig. 6 compares ESRGAN and LDL-SR across the three region types (Type A, B, and C). LDL-SR reduces residual variance in artifact-prone regions while preserving valid detail, with the largest improvements in Type C regions. This behavior aligns with LDL design and explains its effectiveness in suppressing hallucinated artifacts [1].

VI. Discussion and Outlook

Although this study focuses on supervised super-resolution with paired training data, the results show that locally discriminative, variance-aware supervision can effectively reduce hallucinated artifacts while preserving structural fidelity in remote-sensing imagery. While LDL requires high-resolution references during training and introduces modest computational overhead, similar requirements are common in many state-of-the-art super-resolution methods.

A limitation of this study is that per-image evaluation results were not retained for all models, which prevents formal statistical hypothesis testing such as paired t-tests. As a result, the contribution of each component is assessed based on consistent performance trends rather than statistical significance testing.

Future work may explore extending locally discriminative supervision to self-supervised or blind super-resolution settings and integrating LDL-style mechanisms with diffusion-based models [4]. In addition, evaluation on larger remote-sensing datasets and analysis of downstream geospatial tasks may further clarify the practical value of artifact-aware super-resolution for Earth-observation applications.

VII. Conclusion

This paper investigated the use of locally discriminative learning as an artifact-aware supervision strategy for remote-sensing image super-resolution. Motivated by the need for structurally reliable reconstruction in Earth-observation imagery, experiments on the UC Merced Land Use dataset demonstrate consistent improvements over a standard ESRGAN baseline. The LDL-based model achieved measurable gains, with PSNR increasing by 19.3%, SSIM by 31.2%, and LPIPS decreasing by 39.8%, indicating enhanced structural fidelity and reduced high-frequency artifacts.

Comparisons with a non-adversarial variant and a transformer-based reference model show that simply modifying the architecture or removing adversarial loss does not eliminate hallucinations in structured aerial scenes. By selectively constraining unstable regions during training, LDL achieves balanced optimization, preserving meaningful detail while improving stability. Overall, the results support variance-aware supervision as an effective way to enhance the reliability of super-resolved remote-sensing imagery.

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